Felipe **Cybis Pereira**

CURRENTLY POST-DOC IN PHYSICS FOR MEDICINE PARIS

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Research Experience

Physics for Medicine Paris, Inserm, ESPCI Paris, PSL, CNRS — Paris, France

DOCTORAL DEGREE — February 2021- July 2025

- Adviser: Sophie Pezet, PI, and Mickaël Tanter, PI.
- Title: Revealing vascular patterns in the spatial navigation system with functional ultrasound imaging.
- Techniques: **Functional Ultrasound imaging - Spatial navigation - Rodent's brain imaging - Animal behavior and cognition - Python - 3D design conception (CAD) - Ultrafast Ultrasound**

Harvard University, Rogulja Lab at Harvard Medical School — Boston, USA

RESEARCH INTERN — May 2019- August 2019

- Adviser: Alexandra Vaccaro, PhD, and Dragana Rogulja, PI.
- Focus: Insights in sleep deprivation in *Drosophila melanogaster*.
- Techniques: ***Drosophila melanogaster*** **rearing - Immunostaining - Confocal microscopy - Ethological analysis - Survival and Dietary assays - Drosophila Activity Monitoring (DAM) system**

ICONEUS, Real-time portable functional ultrasound small animal neuroimaging — Paris, France

R&D INTERN — July 2018- December 2018

- Adviser: Bruno Osmanski, PhD, and Mickaël Tanter, PI.
- Focus: Code optimization for transcranial multiplane wave ultrasound imaging.
- Techniques: **Plane wave and Multiplane wave ultrasound imaging - Power Doppler imaging - Brain connectivity - MATLAB - Small animal brain imaging**

Teaching experience

ESPCI Paris - PSL University — Paris, France

TEACHER ASSISTANT IN THE PHYSIOLOGY PRACTICAL WORK FOR THE 2ND YEAR STUDENTS — 2022-2023 and 2023-2024

Professor: Thierry Gallopin, PhD.

- Neuroscience-focused practical work: **(1) human EEG (2) pose-estimation and animal tracking (3) human sleep**.

NeuroPSI - Paris-Saclay Institute of Neuroscience — Saclay, France

TEACHER ASSISTANT IN THE MASTERS 2 FOR COMPUTATIONAL NEUROSCIENCES AND NEUROENGINEERING — 2023 and 2024

Couse unit: Methods for measuring and actuating neuronal activity.

Professor: Isabelle Ferezou, PhD.

- Principles on Ultrafast Ultrasound, functional Ultrasound imaging and Ultrasound Localization Microscopy.

Education background

BioMedical Engineering Master — Paris, France

MASTER'S DEGREE IN BIOENGINEERING AND NEUROSCIENCES — September 2020- August 2021

- Scholarship student for the PSL Graduate Program in Life Sciences.

ESPCI Paris - PSL University — Paris, France

MASTER'S DEGREE IN ENGINEERING FOCUSED IN BIOTECHNOLOGY — September 2016- August 2019

- *Michelin Excellency* scholarship student in a double degree program with UFSC University in Brazil.

Universidade Federal de Santa Catarina (UFSC) — Florianópolis, Brazil

BACHELOR'S DEGREE IN CHEMICAL ENGINEERING — March 2014- March 2020

Skills

Programming

- **Scientific Python**: Neuroimaging (Nipy suite, BrainGlobe), Machine learning (scikit-learn), Visualization (VTK, PyVista).
- **General Python Packaging**: Documentation and examples, unit testing, linting and formatting, pre-commit hooks.
- **Version control**: Intermediate to advanced usage of **Git**, **GitHub** and **GitHub Actions**.
- **Prototyping**: Intermediate knowledge in **Arduino** and **Raspberry Pi**.
- Intermediate knowledge on **Rust (and Rust bindings for Python)**, **Lua**, **core-utils** and **shell-scripting**, **JavaScript**.
- Intermediate knowledge on text editing softwares such as LaTeX and **Typst**.

Spoken languages: **English** (fluent), **French** (fluent), **Portuguese** (native), Spanish (beginner)

Hobbies

I like playing **Tennis** (I played a lot while kid/teen), programming and tweaking my own dotfiles.

Publications

- Le Meur-Diebolt, S., **Pereira, F. C.**, Mariani, J.-C., Kliewer, A., Farinha-Ferreira, M., Bertolo, A., Osmanski, B.-F., Lenkei, Z., & Deffieux, T. (2025, June). *Robust Functional Ultrasound Imaging in the Awake and Behaving Brain: A Systematic Framework for Motion Artifact Removal*. Cold Spring Harbor Laboratory. https://doi.org/10.1101/2025.06.16.659882
- Zucker, N., Le Meur-Diebolt, S., **Cybis Pereira, F.**, Baranger, J., Hurvitz, I., Demené, C., Osmanski, B.-F., Ialy-Radio, N., Biran, V., Baud, O., Pezet, S., Deffieux, T., & Tanter, M. (2025). Physio-fUS: A Tissue-Motion Based Method for Heart and Breathing Rate Assessment in Neurofunctional Ultrasound Imaging. *Ebiomedicine*, *112*, 105581. https://doi.org/10.1016/j.ebiom.2025.105581

Posters in International Conferences

- **Pereira, F. C.**, Ialy-Radio, N., Bhattacharya, S., Nouhoum, M., Osmanski, B.-F., Pezet, S., & Tanter, M. (2024). Functional Ultrasound Tools for Automatic Atlas Registration and Chronic Neuroimaging on Naturally Behaving and Sleeping Rats. *Sleep Medicine*, *115*, S409. https://doi.org/10.1016/j.sleep.2023.11.1098
- **Cybis Pereira, F.**, Ialy-Radio, N., Bhattacharya, S., Osmanski, B.-F., Pezet, S., & Tanter, M. Chronic Functional Ultrasound Imaging Combined with Behaviour Tracking on Freely Moving Rats. *Neuroscience 2022*.
- **Cybis Pereira, F.**, Ialy-Radio, N., Bhattacharya, S., Osmanski, B.-F., Pezet, S., & Tanter, M. Chronic Functional Ultrasound Imaging Combined with Behaviour Tracking on Freely Moving Rats. *FENS Forum 2022*.
- Huang, J., Maguelone, F., Vetere, G., Pezet, S., Travert-Jouanneau, Y., **Cybis Pereira, F.**, Melik-Parsadaniantz, S., Amar, L., Bourgeais Rambur, L., & Réaux-Le-Goazigo, A. (2024). Deciphering the Precise C-Fos Connectome of Ocular Pain in Mice. *2024 ARVO Annual Meeting*, *65*, 2637.
- Bourgeais Rambur, L., Travert, Y., **Cybis Pereira, F.**, Huang, J., Baudouin, C., Melik-Parsadaniantz, S., Deffieux, T., Pezet, S., & Réaux-Le-Goazigo, A. (2023). Ultrafast Ultrasound Imaging of the Trigeminal Ganglion and Brain in Mice. *2023 ARVO Annual Meeting*, *64*, 3361.
- **Felipe Cybis Pereira**, Nathalie Ialy-Radio, Soumee Bhattacharya, Bruno-Félix Osmanski, Sophie Pezet, Mickael Tanter. Chronic functional ultrasound imaging combined with behaviour tracking on freely moving rats. *fUSbrain 2022*.